1 Artificial Vision Project

2 Software

The software has a configuration and a set of parameters. The configuration is static: it defines the chosen operators and devices. Parameters are the values subject to tuning and calibration.

2.1 Cross-cutting mechanisms

2.1.1 Configuration

When the system is started the System-Configuration is read, to determine the cameras and the operators. The scheme of the configuration is known in advance.

2.1.1.1 Pipeline Configuration

Each pipeline configuration has an Expected Frame Rate [Hz]

```
SemPointCloutPipelineConf
  ExpectedFrameRate [Hz]
  InputCameraName
  SemSegmConf
      SemSegmOperatorName:
      OutputResolution: px-width, px-height
  StereoMatcherConf
      StereoMatcherOpName
      OutputResolution: px-width, px-height
  PointCloudFusionOp
      PointCloudFusionOpname
```

2.1.2 Parameters

Parameters are de-serialized from a JSON file. The exact scheme of the parameters is not known, the device drivers and the actual operators know it, and they must tolerate when the schema does not match the parameters because some parameter was added. So, parameters are kept as **`SerializedParams`**, that is, a text string or a byte array.
Scheme of the Camera calibration parameters group.

- Camera Calibration (group)
  - Camera name (group)
    - Intrinsic Calibration // Intrinsic parameters and distortion
      - Calibration Images Dir Path
      - Checkerboard-params (params group)
        - Num Horizontal Corners
        - Num Vertical Corners
        - Checker Width [mm]
        - Checker Height [mm]
      - Intrinsic Matrix
      - Distortion Coefficients
      - Re-projection error
    - Extrinsic Calibration (e.g. pose of the camera w.r.t to the World reference)
      - Checkerboard-params (as above, number of corners, checker dimension)
      - Camera Pose
        - Translation
        - Rotation (Rodrigues)

2.1.3 Frame-Operators

The interface allows to retrieve and update its parameters.
Input / Output operators that act on frames.

```
updateParams( ParamsPath, SerializedParams)
```

2.1.4 Pipeline Operators

A pipeline is created from a factory.
A pipeline can receive frames from either a camera frame source, or a recorded video. Example: run pipeline / run pipeline on video. Thus, the frame source is passed to the factory method. The frame source must be accessed by the GUI to execute the play/pause, go next frame, etc.

2.2 Capabilities of the GUI container

2.2.1 Camera Settings capability

- A camera must be selected
- The current camera parameters are applied to the camera
- The user can browse parameters, modify and apply them
- The user can save or discard the modified parameters
- If parameters are saved, they are applied to the system configuration

Note: the camera itself does not update the parameters, in other words, the camera does not know the parameters path, because it can depend on the situation.

2.2.2 Camera Focus capability

The camera focus acts on an image of a camera.

- A camera must be selected
- The current camera parameters are applied to the camera

The Camera Focus is a FrameCostOp: it determines the cost of a frame. The output of the FrameCostOp is a subscription to an Entity Cost; the Entity Cost has a range and a value.

2.2.3 Camera Video Acquisition capability

- A camera must be selected
- For each image stream, a separate directory is created
- It cannot record more than one camera: it is related to a physical, named camera.

2.2.4 Camera Calib. Image Acquisition capability

This capability is based on the Checkerboard2D Ft.Points Operator.

- A camera must be selected
- The current camera settings are applied
- The Checkerboard 2D Ft.Points operator has a frame as input and for each image, it produces a vector of pairs (x,y);(u,v), where x,y is the corner coordinate on the checker and (u,v) the corresponding coordinate on the image.
- If the vector is empty it means that the operator was not able to retrieve the corners; otherwise the images can be saved.
- The operator has also a frame output

2.2.5 Mono Camera Intrinsic Calibration (capability)

It is a one-shot operator.
A Camera must be selected
MonoCameraIntrinsicCalibOp: an operator that has, as input, a frame iterator where the total number of frames is known.
The operator can receive the serialized parameters, the path (e.g. Camera Calib / Camera name / Intrinsic Calib), to update its internal parameters.
Then, it is possible to export the parameters.
The GUI can inspect the parameters, the usual way.
The GUI can run the one-shot operator.
The result is an IntrinsicCalib params (future)
The result can be serialized back among the parameters

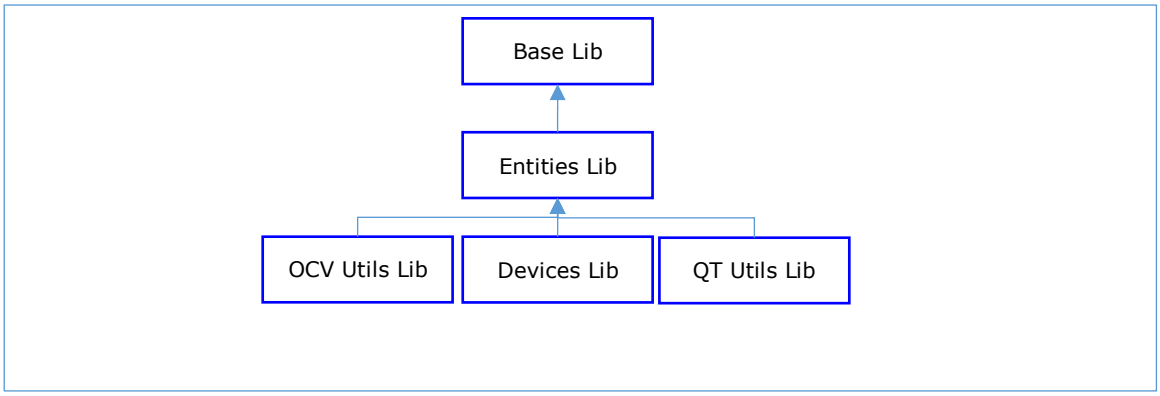
2.2.6 Stereo Camera Intrinsic Calibration (capability)

- It is a one-shot operator
- A camera must be selected
- The input is a frame iterator: the total number of frames is known in advance.
- The result is a StereoCameraCalibParams object (a future), that is saved back among the parameters.

2.3 Software Architecture

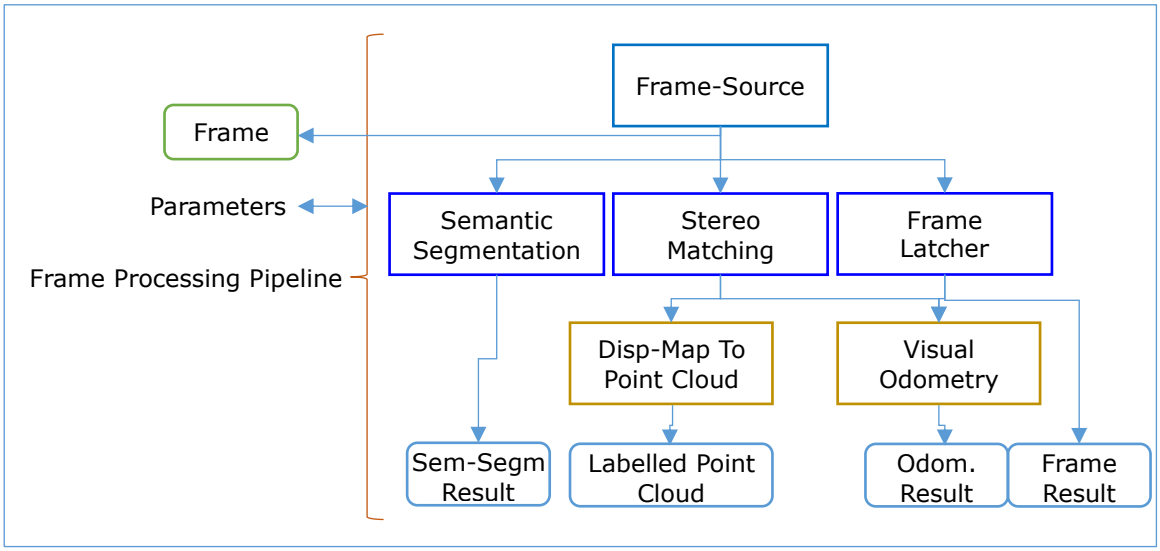
As per Simon Brown, the hierarchy is: Context, Containers, Components, Classes.

2.3.1 Dependencies between libraries and containers



- Base Lib: Tracing, patterns, actors
- Entities Lib: data-classes and problem domain classes, without algorithms: configuration, image/frame, etc.
- Devices Lib: devices management e.g. cameras
- OCV Utils Lib: OpenCV utils library
- QT Utils Lib.

2.3.2 Architecture of the frame processing pipeline



2.3.2.1 Frame Operators

Each frame operator must have, as output a **frame-referenced result**; the resulting class type must derive from frame-referenced, it means that it is possible to retrieve the related frame number. Example, the frame is frame-referenced.
This way, an operator receiving multiple inputs can run when all inputs match with regard to the referenced frame.

2.3.2.2 Pipeline

A pipeline has:

- A set of named operators
- From each operator is it possible to retrieve and update its frames
- It is possible to iterate all operators and save their parameters under for example:
    - Pipeline operators (group)
        - Camera Name (group)
            - SemSegm operator (group)
                - Parameters …..

- SGBM Stereo Matcher Op
  - Parameters……

Depending on its type the pipeline has a set of subscriber result, referred to a frame

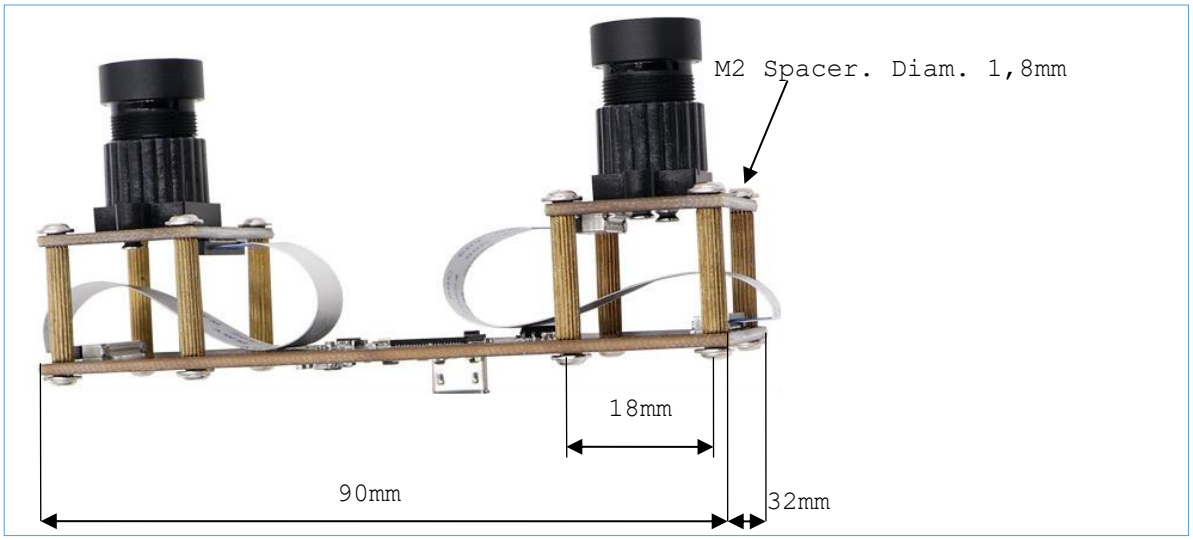
2.3.2.3 Label Point Cloud pipeline

A pipeline that has as output:

- The latched frame (frame referenced)
- A vector of labelled points; for each item (X;Y;Z,bitmask), referred to the world
- The result of the semantic segmentation

3 Mechanical

3.1 How to mount the Camera KYT-U200-SNF01



Calculation of the camera baseline:

Input	Input	Input	Input	Input	Input	Input	Input	Input	Result		Input	Result
Image Width [px]	Image Height [px]	Image Ratio	Image Width [m]	Image Height [m]	Pixel Width [m]	Pixel Height [m]	Focal Length [m]	Max Z [m]	Baseline [mm]		Baseline [mm]	Max Z [m]
1520	1080	1,41	0,0054	0,0031	3,58E-06	2,84E-06	0,0021	30	51,13		60,00	44,30
1280	960	1,33	0,0054	0,0031	4,25E-06	3,20E-06	0,0021	30	60,71		60,00	39,38
1280	720	1,78	0,0054	0,0031	4,25E-06	4,27E-06	0,0021	30	60,71		60,00	29,53
800	600	1,33	0,0054	0,0031	6,80E-06	5,12E-06	0,0021	30	97,14		60,00	24,61
640	480	1,33	0,0054	0,0031	8,50E-06	6,40E-06	0,0021	30	121,43		60,00	19,69

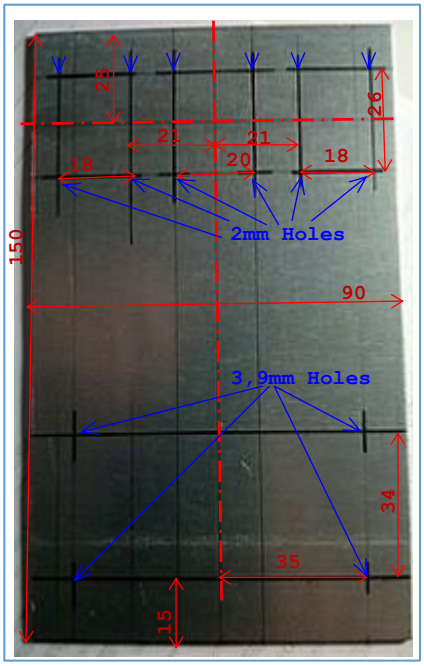
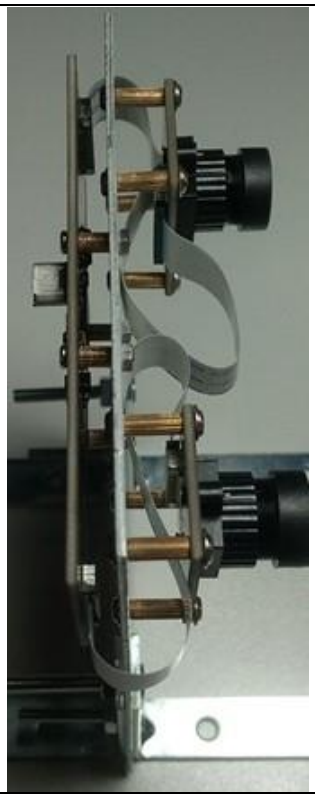


Figure 1 - The sheet metal holding the camera

Measures are expressed into millimeters.





4 User Interface

The icons used in the QT GUI program are part of the icon-set called "Pretty Office 8"; they were downloaded from the link:

- https://www.iconarchive.com/show/pretty-office-8-icons-by-custom-icon-design.html

5 System Integration and setup

5.1 Camera Calibration

In the JSON Camera Configuration section, each camera has a URI parameter expressed as Scheme (e.g. OCV) and a list of key-value pairs. The set of allowed keys depend on the driver. Example:

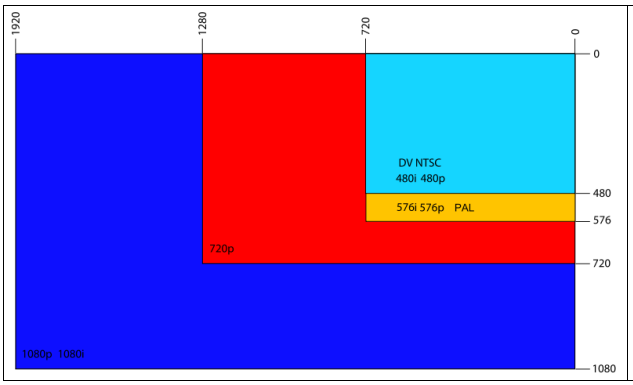
```
    "CameraConfig": [
      {
        "Id": "BuiltInCam",
        "URI": "ocv://?index_main=0;res_width=640;res_height=480;frame_rate=20",
        "DriverId": "OcvCamera",
        "Enabled": false
      },
```

Example of calibration, with valid camera resolutions:

```
{
  "DeviceConfig": {
    "CameraConfig": [
      {
        "Id": "BuiltInCam",
        "URI": "ocv://?index_main=0;res_width=640;res_height=480;frame_rate=20",
        "DriverId": "OcvCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U130-Camera-ResMax",
        "URI": "ocv://?index_main=1;res_width=2560;res_height=960;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U130-Camera-ResMin",
        "URI": "ocv://?index_main=1;res_width=1280;res_height=480;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U200-Camera-Res0",
        "URI": "ocv://?index_main=1;res_width=1280;res_height=480;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U200-Camera-Res1",
        "URI": "ocv://?index_main=1;res_width=1600;res_height=600;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U200-Camera-Res2",
        "URI": "ocv://?index_main=1;res_width=2560;res_height=720;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U200-Camera-Res3",
        "URI": "ocv://?index_main=1;res_width=2560;res_height=960;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Kyt-U200-Camera-Res4-MAX",
        "URI": "ocv://?index_main=1;res_width=3040;res_height=1080;frame_rate=30",
        "DriverId": "OcvDualLensCamera",
        "Enabled": false
      },
      {
        "Id": "Minoru-Camera",
        "URI": "ocv://?index_main=1;res_width=320;res_height=240;frame_rate=20",
        "DriverId": "OcvMultiCamera",
        "Enabled": false
      }
    ]
  }
}
```

6 Reference

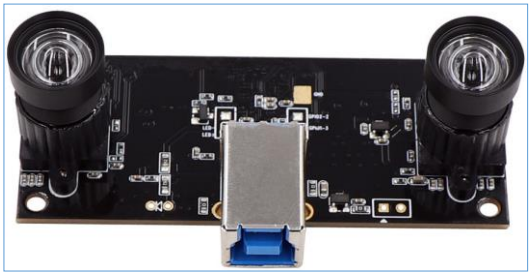
6.1.1 Video Resolutions



- 640x480
- 1280x720 = HD Ready or HD; 16:9
- 1920x1080 = Full HD; 16:9

7 Datasheets

7.1 Camera Kayeton Dual Lens KYT-U130-3R60



USB VendorID ProductID = 2304:1689 (HEX)
Feature:
*Synchronization 1.3MP HD 960P Stereo 3D VR Dual Lens USB3.0 Camera Module
  with Lens 2.1mm(2.8mm/3.6mm/6mm/8mm/12mm/16mm, fisheye, non distortion optional)
*Synchronization technology
*High speed USB 3.0 interface
*High quality image
*Low power consumption
*Small outline for embedded application
*UVC compliance for use in OS Windows, Linux, Android, Mac
*OTG support

Specification:
Model: KYT-U130-3R60
Sensor: Aptina AR0130
Focus: Manual
Synchronization stereo camera: Same frame rate and same resolution at same time for each camera
Resolution & frame :
(2×1280)x960 MJPEG 30fps, YUY2 30fps
(2×640)x480 MJPEG 30fps, YUY2 30fps
Compression format: MJPEG \ YUY2
Interface: USB3.0
Lens Parameter : 2.1mm(2.8mm/3.6mm/6mm/8mm/12mm/16mm, fisheye, non distortion optional)
Voltage: DC5V
USB Video Class(UVC): Support
OTG: Support
Auto exposure AEC: Support
Auto white blance AEB: Support
Adjustable parameters: Brightness/Contrast/Color saturation/Definition/Gamma/WB
Dimension: Board 74mm(L)x30mm(W),Sensor distance 60mm,other size can be customize
Operating Temperature: -20°C to 70°C
Support OS: Windows, Linux, MAC, Android
Link

7.2 Camera Kayeton Dual Lens KYT-U200-SNF01



USB VendorID ProductID = 2048:2088 (HEX)
Feature:
*Synchronization 2Megapixel HD 1080P Stereo USB2.0 Camera Module with Flexible Cable
  with Lens 2.1mm(2.8mm/3.6mm/6mm/8mm/12mm/16mm, fisheye, non distortion optional)
*Synchronization technology
*High speed USB 2.0 interface
*High quality image
*Low power consumption
*Small outline for embedded application
*UVC compliance for use in OS Windows, Linux, Android, Mac
*OTG support

Specification:
Model: KYT-U200-SNF01

Sensor: OmniVision OV4689 (see 7.3 “Omnivision OV4689 – Image Sensor”)
Focus: Manual
Synchronization: Yes
Resolution & frame :
(2×1520)x1080 MJPEG 30fps,YUY2 1fps
(2×1280)x960 MJPEG 30fps,YUY2 1fps
(2×1280)x720 MJPEG 30fps,YUY2 1fps
(2×800)x600 MJPEG 30fps,YUY2 3fps
(2×640)x480 MJPEG 30fps,YUY2 5fps
(2×1088)x1520 MJPEG 30fps,YUY2 1fps
(2×960)x1280 MJPEG 30fps,YUY2 1fps
(2×720)x1280 MJPEG 30fps,YUY2 3fps
(2×600)x800 MJPEG 30fps,YUY2 3fps
(2×480)x640 MJPEG 30fps,YUY2 5fps
Compression format: MJPEG \ YUY2
Interface: USB2.0
Lens Parameter : 2.1mm(2.8mm/3.6mm/6mm/8mm/12mm/16mm, fisheye, non distortion optional)
Voltage: DC5V
USB Video Class(UVC): Support
OTG: Support
Auto exposure AEC: Support
Auto white blance AEB: Support
Adjustable parameters: Brightness/Contrast/Color saturation/Definition/Gamma/WB
Dimension: Board 90mm(L)x30mm(W),Cable 100mm(L),other size can be customized
Operating Temperature: -20°C to 70°C
Support OS: Windows, Linux, MAC, Android
Link
The driver is based on the Linux UVC library ([UVC05]), with the OpenCV Video for Linux driver, it took about 100ms to retrieve an image; with the UVC library it takes about 5ms.

7.3 Omnivision OV4689 – Image Sensor

Link
Pixel Size: 2.0 µm x 2.0 µm
Active Array Size: 2688 x 1520
Image Area: 5440 x 3072 µm
Package Dimensions: 6630 x 5830 µm
Shutter Type: Rolling Shutter
Maximum image transfer rate: 2688 x 1520 @90fps

7.4 Monoru 3D WebCam

Specifications:
VGA CMOS Sensor
Maximum resolution 800×600
Camera baseline length 6 cm
Maximum frame rate 30 fps
L/R shutters don’t synchronize(Max deviation 16.5 ms)
Manual focus(from 10 cm to infinity)
field angle 42 degrees (actual measurement)

Notes
Indoor use only since the aperture is always open. To use outdoors, suppress the amount of light using appropriate filters.
Each webcam is recognized as “Vimicro USB2.0 UVC PC Camera”.
At this stage it is equivalent to having two discrete webcams.
With Minoru driver in the CD included, the PC recognizes the device as “Novo – Minoru”. This device outputs the anaglyph or other 3d video and can be used for applications like MSN Messenger, Skype, etc.
Calibration and rectification is required. Adjusting vertical alignment only is sufficient in most cases. Lense distortion is not affecting level
https://www.minoru3d.com/



8 Reference documentation

ID	Document	Version
[UVC05]	“Universal Serial Bus Device Class Definition for Video Devices”	Rev.1.1, June 1,2005